Problem 1 – No Spurious Local (30 points)

Let $A \in \mathbb{R}^{d \times d}$ be a strictly positive definite matrix and unique eigenvalues $\lambda_1 > \cdots > \lambda_d$. Consider the task of minimizing the objective $\mathcal{L}(w) = \frac{1}{4}\|A - ww^\top\|_F^2$.

a) Find the gradient of $\mathcal{L}(w)$.

b) Find all of the first-order stationary points of $\mathcal{L}(w)$.

c) Note that $\nabla^2\mathcal{L}(w) = 2ww^\top - A + \|w\|^2 I_d$. Prove that $\lambda_{\min}(\nabla^2\mathcal{L}(w)) < 0$ for all first-order stationary points that are not a global minimum.

d) Using the previous result find all of the global minima.

Problem 2 – Generalizing PL (30 points)

In this problem we consider functions that satisfy the following α-convexity property

$$f(y) \geq f(x) + \langle \alpha(y, x), \nabla f(x) \rangle \tag{1}$$

for some vector-valued function $\alpha(y, x) : \mathbb{R}^{d \times d} \to \mathbb{R}^d$. Note that with $\alpha(y, x) = y - x$ we recover the usual definition of convexity.

a) Assume f does not have any first-order stationary point. Show that f is α-convex for some $\alpha(y, x)$.

b) Assume f is α-convex. Show that if it has any stationary point, then that point is also a global minimum of f.

c) Assume f satisfies the PL inequality/condition with constant μ. Show that f is α-convex.

Problem 2 – Convergence of GAN training algorithms (40 points)

In this problem consider a simple GAN training example. Please submit your code as part of the solution.

Consider GAN with Discriminator $D_y(z) = y^\top z$ and generator $G_x(Z) = Z + X$. Assume we are interested to learn a two dimensional distribution $q = \mathcal{N}([3,4]^\top, I_2)$.

a) Using the above construction, write down precisely what the minmax problem is for WGAN training

$$\min_x \max_y U(x,y) = \mathbb{E}_{z\sim q}[D_y(z)] - \mathbb{E}_{Z\sim\mathcal{N}(0,I)}[D_y(G_x(Z))] \tag{2}$$

by calculating a closed form expression for the expectations.

b) Argue that $U(x,y)$ is concave in y for any x and convex in x for any y

c) Find the equilibrium point, i.e. (x^*, y^*) such that for any x, y

$$U(x^*,y^*) \geq U(x^*,y), \qquad U(x,y^*) \geq U(x^*,y^*) \tag{3}$$

(implying $p_{G_{x^*}} = q$)

d) Initialize $y = [1,1]^\top$ and $x = [1,1]^\top$. Implement the GD, Proximal point, EG, and OGD methods for this task and report the learning rates you used (set $T = 100$).

e) Repeat the last part expect now for the case where we replace the expectations with sample averages. Notably, sample $N = 100$ points from q ($z_i \sim q$) and sample $N = 100$ points from $\mathcal{N}(0, I_2)$ ($Z_i \sim \mathcal{N}(0, I_2)$) and solve

$$\min_x \max_y \hat{U}(x,y) = \frac{1}{N}\sum_{i=1}^N D_y(z_i) - \frac{1}{N}\sum_{i=1}^N D_y(G_x(Z_i)) \tag{4}$$

using the GD, Proximal point, EG, and OGD methods for this task and report the learning rates you used (set $T = 100$).